



# **Understanding H<sub>2</sub>S Safety**

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# What is Working at Height

Hydrogen Sulfide ( $H_2S$ ) is a toxic and explosive gas found naturally in oil and gas fields. It is also found where decaying organic matter is found, such as in swamps and marshes, garbage dumps and sewage pipes. It is also found naturally in volcanoes, underwater thermal vent sand hot springs.





# H<sub>2</sub>S and the Sense of Smell

- Hydrogen sulfide (H<sub>2</sub>S) has a strong “rotten egg” smell at low concentrations.
- However, At higher concentrations, you cannot smell hydrogen sulfide because it destroys the sense of smell.
- After short exposure, you can no longer detect its odor, even if the gas concentration increases.
- This loss of smell creates a false sense of safety.
- You might think the gas has disappeared, but the concentration could be rising to dangerous or deadly levels.





# Important Properties of H<sub>2</sub>S

|                             |                                     |
|-----------------------------|-------------------------------------|
| <b>Molecular Weight:</b>    | 34 kg/kmole                         |
| <b>Gas Density:</b>         | 1.93 kg/m <sup>3</sup>              |
| <b>Specific Gravity:</b>    | 1.189 (heavier than air)            |
| <b>Solubility in Water:</b> | 4.67 m <sup>3</sup> /m <sup>3</sup> |
| <b>Flammability Limits:</b> | 4.3% to 45.5% by volume in air      |
| <b>TLV value</b>            | 1 ppm (TWA) and 5 ppm (STEL)        |

- TLV (Threshold Limit Value): How much H<sub>2</sub>S a worker can be safely exposed to on average during an 8-hour workday and a 40-hour workweek.
- For Hydrogen Sulfide (H<sub>2</sub>S):
  - TWA (Time Weighted Average): 10 ppm ( 8 hour shift)
  - STEL (Short-Term Exposure Limit): 15 ppm



# Important Properties of H<sub>2</sub>S

$$\text{Adjusted TLV} = \text{TLV} \times \left( \frac{40}{h_w} \right) \times \left( \frac{168 - h_w}{128} \right)$$

where  $h_w$  = total hours worked per week.

For a 12-hour × 7 days schedule:

$h_w = 84$  hours per week

$$\text{For } h_w = 84 : \quad \text{Adjusted TLV} = 10 \times \left( \frac{40}{84} \right) \times \left( \frac{168 - 84}{128} \right) = 3.125 \text{ ppm}$$





# Hazard vs Risk – The Difference

- People often use “hazard” and “risk” interchangeably, but they are different.
- **Hazard:** An inherent property of a material that could cause harm.
  - Example: A tank full of gasoline can catch fire.
- **Risk:** The probability that the hazard will actually cause harm or damage.
  - Example: The chance that the gasoline tank catches fire and causes damage



# Hazard and Risk in H<sub>2</sub>S

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## ➤ Hazard:

- The presence of Hydrogen Sulfide (H<sub>2</sub>S) gas in a facility or work environment.
- It is toxic, flammable, and heavier than air, making it inherently dangerous.

## ➤ Risk:

- The likelihood that the hazard may lead to harmful events, such as:
  - Gas leaks
  - Fire or explosions
  - Health effects on personnel (e.g., respiratory problems, unconsciousness, or death)
  - Damage to equipment, infrastructure, and the facility
  - Environmental contamination



# The Hidden Danger

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| H <sub>2</sub> S PROPERTY                      | HAZARD                                                                    |
|------------------------------------------------|---------------------------------------------------------------------------|
| Colorless                                      | Invisible: cannot be seen                                                 |
| Toxic                                          | Lethal in small doses                                                     |
| Odor at low concentrations                     | Destroys sense of smell as concentration increases                        |
| Heavier than air                               | Collects in low lying areas                                               |
| Gaseous form                                   | Moves with the wind for KMs and easily inhaled                            |
| Soluble in Oil and Water                       | Forms sulfuric acid (H <sub>2</sub> SO <sub>4</sub> ), which is corrosive |
| Reacts with metals to form sulfides and oxides | Forms iron sulfide (FeS) which can ignite when exposed to air             |





# Risk Equation in Industry

- In industrial safety, risk is often calculated using the Risk Equation:

**Risk= Probability of Occurrence X Severity of Consequences**

| <b>Reducing Probability of Exposure:</b>                                  | <b>Reducing Severity of Consequences:</b>                   |
|---------------------------------------------------------------------------|-------------------------------------------------------------|
| Continuous H <sub>2</sub> S monitoring with gas detectors                 | Use of personal protective equipment (PPE) like respirators |
| Proper ventilation in confined or low-lying areas                         | Emergency response plans and evacuation procedures          |
| Safe work practices and controlled access to H <sub>2</sub> S-prone areas | Explosion-proof equipment in H <sub>2</sub> S areas         |
| Regular inspection and maintenance of equipment                           | Training workers on first aid and H <sub>2</sub> S hazards  |



# Effects of H<sub>2</sub>S on Health

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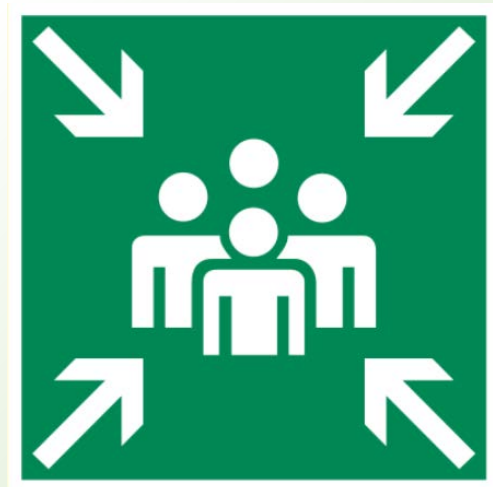
| H <sub>2</sub> S Concentration | Effects on typical human                                                                                                              |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 0-10 ppm                       | Irritation of eyes, nose and throat                                                                                                   |
| 10-50 ppm                      | Headache<br>Dizziness<br>Nausea and vomiting<br>Coughing and breathing difficulty                                                     |
| > 50 ppm                       | Severe respiratory tract irritation<br>Eye irritation / acute conjunctivitis<br>Shock<br>Convulsions<br>Coma<br>Death in severe cases |



# Assembly Points for H<sub>2</sub>S

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- Hydrogen Sulfide (H<sub>2</sub>S) is heavier than air.
- In case of a leak, the gas tends to accumulate in low-lying areas.
- Safety Measure:
  - Ensure that emergency Assembly Points are located at elevated locations.
  - This prevents workers from being exposed to H<sub>2</sub>S during evacuation or emergencies.





# H<sub>2</sub>S Detection

- Every rig is required to have the following types of detectors for H<sub>2</sub>S and other toxic and flammable substances:



**Fixed Detector**



**Portable Detector**



**Personal Gas Monitor**

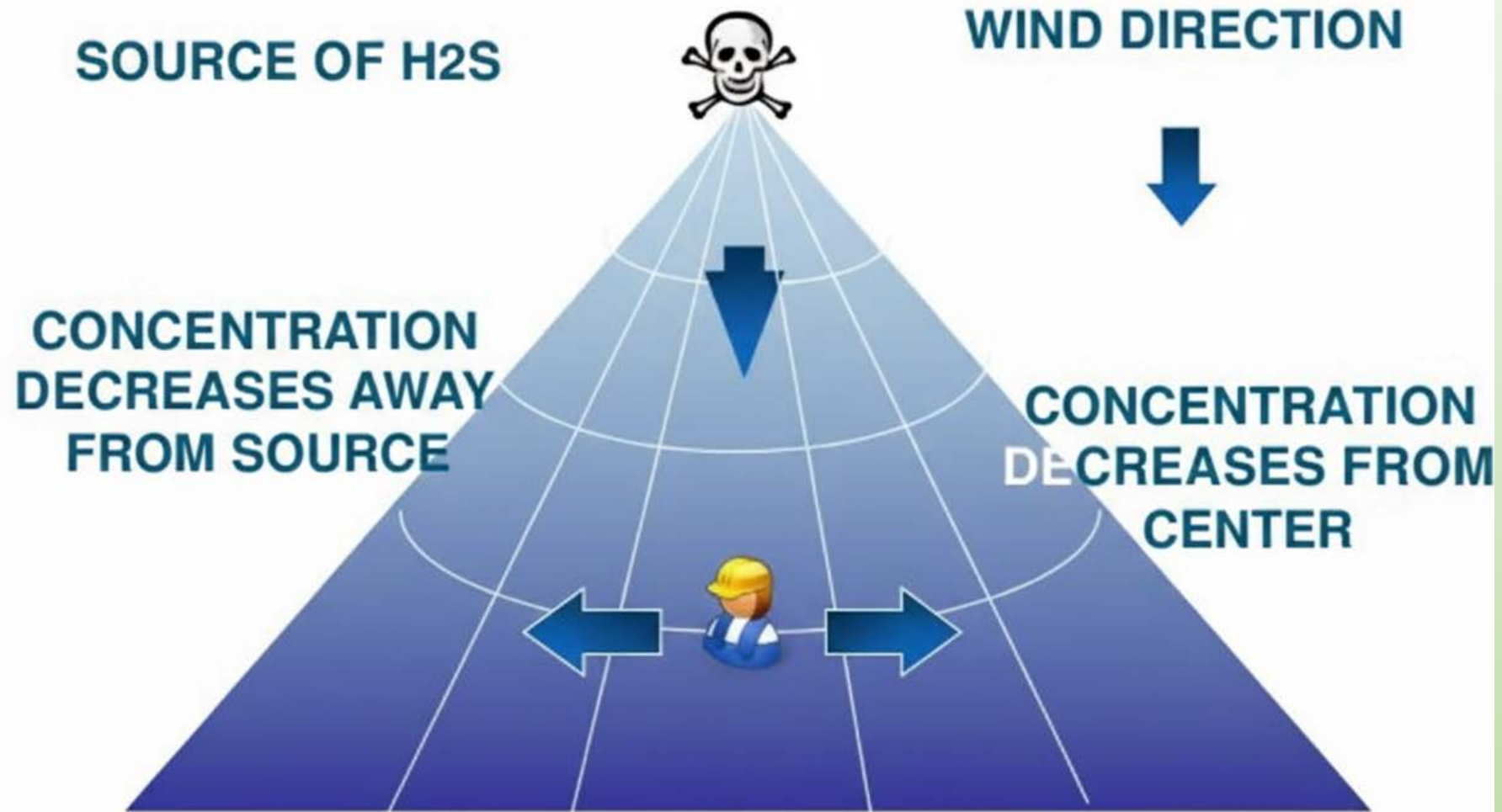


# Emergency Response Plans

- Rig sites must have at least two designated safe briefing areas (SBA) or muster stations
- One safe briefing area must be positioned upwind from potential H2S release points at all times
- Remain calm, don't panic.
- Note the wind direction by checking the nearest windsock
- All personnel report to the upwind designated SBA when an H2S gas release occurs
- Moving crosswind, quickly make your way to the appropriate Safe Briefing Area
- A headcount is completed as per the Station Bill or persons onboard list (POB).



# WHICH WAY TO GO



**GO CROSS-WIND OR UP WIND**





# Hierarchy of Controls

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## Engineering controls

- Scrubbers, poor boy degassers, maintaining two downhole barriers at all times, redundancy, H<sub>2</sub>S rated steel.
- Ventilation – Forced air in pits, bug blowers to push or blow H<sub>2</sub>S in opposite direction
- Chemicals – Reducing H<sub>2</sub>S levels in mud by introducing H<sub>2</sub>S scavenger to drilling fluid
- Flare Pits – Divert H<sub>2</sub>S away from the rig and prevent gas spreading by burning off the H<sub>2</sub>S



# Safe Work Practices

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- Only smoke in designated areas
- Inspect your SCBA.
- Obey posted signs & barriers
- Always be aware of the wind direction
- Always have two escape routes from your work area
- Always follow Permit to Work (PTW) & Log-Out, Tag-Out (LOTO) procedures



# PPE (SCBA)

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- Carries its own compressed air cylinder
- For environments immediately dangerous to life or health (IDLH), e.g., high H<sub>2</sub>S concentration, smoke, or confined spaces.
- Contains 30 minutes of air approximately
- Emergency, short-term, fully portable



# PPE (SABA)

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- Air is supplied through a hose from a remote air supply
- Longer-term work in hazardous areas, air supplied from a safe source.
- As long as air is supplied
- Limited by hose length